

Tatsuya Terao

DOCTORAL STUDENT

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Research Interests

- Theoretical Computer Science.

Education

Kyoto University

BACHELOR OF SCIENCE

- Faculty of Science, Division of Physics

Kyoto, Japan

April 1, 2018 - March 31, 2022

Kyoto University

MASTER OF SCIENCE

- Advisor: Prof. Yusuke Kobayashi

Kyoto, Japan

April 1, 2022 - March 31, 2024

Kyoto University

DOCTOR OF SCIENCE

- Advisor: Prof. Yusuke Kobayashi

Kyoto, Japan

April 1, 2024 - present

Professional Experience

2024-2027 Research Fellowships for Young Scientists (DC1), Japan Society for the Promotion of Science

Publications

Authors are listed alphabetically. Exceptions are marked with †.

1. Yusuke Kobayashi and Tatsuya Terao: One-Face Shortest Disjoint Paths with a Deviation Terminal,
In Proceedings of the 33rd International Symposium on Algorithms and Computation (**ISAAC 2022**), 47:1–47:15.
[doi:10.4230/LIPIcs.ISAAC.2022.47](#)
2. Tatsuya Terao: Faster Matroid Partition Algorithms,
In ACM Transactions on Algorithms (**TALG**), Volume 21, Issue 2, 2025.
[doi:10.1145/3707208](#)
A preliminary version appeared in Proceedings of the 50th EATCS International Colloquium on Automata, Languages and Programming (**ICALP 2023**), 104:1–104:20.
[doi:10.4230/LIPIcs.ICALP.2023.104](#)
3. Yusuke Kobayashi and Tatsuya Terao: Subquadratic Submodular Maximization with a General Matroid Constraint,
In Proceedings of the 51st EATCS International Colloquium on Automata, Languages and Programming (**ICALP 2024**), 100:1–100:19.
[doi:10.4230/LIPIcs.ICALP.2024.100](#)
4. Tatsuya Terao and Ryuhei Mori: Parameterized Quantum Query Algorithms for Graph Problems †,
In Proceedings of the 32nd Annual European Symposium on Algorithms (**ESA 2024**), 99:1–99:16.
[doi:10.4230/LIPIcs.ESA.2024.99](#)
5. Tatsuya Terao: Deterministic $(2/3 - \epsilon)$ -Approximation of Matroid Intersection Using Nearly-Linear Independence-Oracle Queries,
In Proceedings of the 19th Algorithms and Data Structures Symposium (**WADS 2025**), 50:1–50:18.
[doi:10.4230/LIPIcs.WADS.2025.50](#)
6. Chien-Chung Huang, Naonori Kakimura, Yusuke Kobayashi, Tatsuya Terao: Polynomial Kernels with Reachability for Weighted d -Matroid Intersection,
To appear in Proceedings of the 27th Conference on Integer Programming and Combinatorial Optimization (**IPCO 2026**).

7. Tatsuya Terao: Faster Approximate Linear Matroid Intersection,
To appear in Proceedings of the 20th Scandinavian Symposium on Algorithm Theory (**SWAT 2026**)

Presentations

- One-Face Shortest Disjoint Paths with a Deviation Terminal.
 - ISAAC 2022, Seoul, Korea, Dec. 20, 2022.
- Faster Matroid Partition Algorithms.
 - ICALP 2023, Paderborn, Germany, July 14, 2023.
 - 13th Hungarian-Japanese Symposium on Discrete Mathematics and Its Applications, Tokyo, Japan, May 26, 2025.
- Subquadratic Submodular Maximization with a General Matroid Constraint.
 - ICALP 2024, Tallinn, Estonia, July 9, 2024.
- Parameterized Quantum Query Algorithms for Graph Problems.
 - ESA 2024, Egham, United Kingdom, Sep. 4, 2024.
- Deterministic $(2/3 - \varepsilon)$ -Approximation of Matroid Intersection Using Nearly-Linear Independence-Oracle Queries
 - WADS 2025, Toronto, Canada, Aug. 13, 2025.

Services

PEER REVIEW

I've been a subreviewer for the conference: RANDOM 2025, IPCO 2026, ICALP 2026.